MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSEECT200 IntroductiontoEmbeddedSystem

Duration:2hours **MaximumMarks:70**

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest (I)andAppreciation(Ap)*

Part A

MultipleChoiceQuestions

Answer**All**Questions,Eachquestioncarries1 mark

1	Whatistheprimarypurposeofanembedded system?	U	1
	A. Generalcomputing B.Web browsing		
	C.Performingaspecifictask D.Dataanalysis		
2	Howmanygeneral-purposeregistersdoestheAVRarchitectureprovide?	U	2
	A.8 B.16 C.32 D.64		
3	WhyisEmbeddedCpreferredforembeddedsystemsprogramming?	U	3
	A. ItrunsfasterthanC++		
	B. Itsupportsobject-orientedfeatures		
	C. Itisoptimizedforhardwarecontrolandreal-timeperformance		
	D. Itrequiresnocompiler		
4	Whichregisterpairistypicallyusedtostorea16-bitresultinAVRassembly?	K	4
	A. R0–R1 B.R24–R25 C.R10–R11 D.R16–R17		
5	Whicharchitectureusesseparatememoryforprogramanddata?	U	1
	A.Von Neumann B.RISC C.Harvard D.CISC		
6	Whichinstructionisusedtomoveimmediatedataintoaregister inAVR?	U	2
	A.MOV B.LDI C.LDS D.ST		
7	Whichdatatypeismostsuitabletostorean8-bitvalueinEmbeddedC?	A	3
	A.int B.float C.char D.uint8_t		
8	Which instruction can be used for checking if a number is even or odd in AVR assembly?	A	4
	A. ANDI B.LDI C.ROR D.SUB		
9	Inanembeddedsystem,whatdoesamicrocontrollertypicallyinclude?	U	1
	A. OnlyaCPU B.CPU,RAM,ROM,andI/Oports		
	C.OnlyROM D.Onlyinput/outputports		
10	Whatisthetotal numberofI/OpinsavailableinATMEGA32?	U	2
	A.16 B.24 C.32 D.40		
11	InAVREmbeddedC,whatdoesDDRB=0xFF;do?	A	3
	A. SetsPORTBasinput B.SetsPORTBasoutput		
	C.ReadsfromPORTB D.Delaystheprogram		
12	Whichofthefollowingdisplaydevicesneedsadecodertodisplaynumericdigits?	A	4
	A.LCD B.LED C.SevenSegment Display D.Buzzer		
13	Themainprocessorinanembeddedsystemistypicallya:	U	1
	A. GPU B.MicroprocessororMicrocontroller		
	C.HardDisk D.DSPprocessor		
14	Whichofthefollowingregistersstoresprocessorstatusflags?	U	2
	A.R16 B.SREG C.PC D.SP		

15	Which timer mode allows counting external events in ATmega32? A.PWMModeB.CTCMode C.Normal Mode D.Counter Mode	K	3
16	Which of the following is required to turn ON an LED connected to PORTB.0 using Embedded C? A. PORTB&= ~(1<<PB0); B.PORTB|=(1<<PB0); C.PORTB^=(1<<PB0); D.PORTB=0x00;	A	4
17	WhichofthefollowingisnottypicallyfoundinanembeddedsystemA. ROM B.RAM C.OperatingSystem D.ExternalGPU	U	1
18	TheinstructionROLperformswhichofthefollowingoperations? A. RotaterightthroughcarryB.Shiftrightlogically C.RotateleftthroughcarryD.Complementregister	K	2
19	WhichregisterinAVRmicrocontrollerisresponsibleforenablingglobalinterrupts? A.TIMSK B.GICR C.SREG D.MCUCR	A	3
20	WhatdoesAVRstandforinAVRmicrocontroller? A. AdvancedVirtual RISC B.AtmelVirtualRISC C.AutomaticVoltageRegulator D.AdvancedVLSIRISC	K	1

1 * 20 = 20

Part B

ShortAnswer Questions

Answerany**6**Questions,Eachquestioncarries5marks

21	StatemajorfeaturesofRISCarchitecture	K	1
22	Differentiate between general-purpose registers and special function registers in AVR.	U	2
23	ListanyfourcommonlyuseddatatypesinEmbeddedCandspecifytheirbitsizes.	K	3
24	WriteanAVRassemblycodefor8-bitaddition	A	4
25	Bringoutthereasonsfortheusagemicrocontrollersinembeddedsystems	U	1
26	WhatistheroleoftheflagregisterinATMEGA32?Nameanytwoflags.	U	2
27	MentionthethreebasicsectionsofatypicalAVREmbeddedCprogram.	K	3
28	WriteanEmbeddedCcodetoblinkanLEDconnectedtoPORTD.0.	A	4

6 * 5 = 30

Part C

Essay Questions

Answerany**2**Questions,Eachquestioncarries10marks

29	Discuss the differences between CISC and RISC architectures. Explain how RISC architecture is most recommended for embedded system applications.	U	1
30	ExplainthearchitectureandkeyfeaturesoftheATMEGA32microcontroller.	K	2
31	Discuss the importance of Embedded C in microcontroller-based systems. Describe the structure of an AVR Embedded C program, and explain how timers/counters are handledwithcode examples.	U	3
32	Write an AVR assembly routine to count the number of ones in a byte. Also check if a number is even or odd in assembly language?	A	4

2 * 10 = 20